Introduction

In the twenty years since the first edition of this encyclopedia in 1996, herbal medicine has gone through unprecedented change. Herbs, which have always been the principal form of medicine in developing countries, have again become popular in the developed world, as people strive to stay healthy in the face of chronic stress and pollution, and to treat illness with medicines that work in concert with the body's defenses. A quiet revolution has been taking place. Tens of millions of people now take herbs such as ginkgo (*Ginkgo biloba*, p. 100) to help maintain mental and physical health, and increasingly people consult trained herbal professionals and naturopaths for chronic or routine health problems. Increasingly too, scientific evidence is accumulating to show that herbal medicines can provide treatment that is as effective as conventional medicines but with few side effects. Sales of herbal medicines continue to grow year after year—by over 50 percent in the U.S. since 2000—and several mainstream pharmaceutical companies now manufacture and market herbal medicines.

Plant Medicines

The variety and sheer number of plants with therapeutic properties are quite astonishing. Some 50,000 to 70,000 plant species, from lichens to towering trees, have been used at one time or another for medicinal purposes. Today, Western herbal medicine still makes use of hundreds of native European plants, as well as many hundreds of species from other continents. In Ayurveda (traditional Indian medicine) about 2,000 plant species are considered to have medicinal value, while the Chinese *Pharmacopoeia* lists over 5,700 traditional medicines, mostly of plant origin.

About 500 herbs are still employed within conventional medicine, although whole plants are rarely used. In general, the herbs provide the starting material for the isolation or synthesis of conventional drugs. Digoxin, for example, which is used for heart failure, was isolated from common foxglove (*Digitalis purpurea*, p. 202), and the contraceptive pill was synthesized from constituents found in wild yam (*Dioscorea villosa*, p. 91).

Ecological Factors

The increased use of medicinal herbs has important environmental implications. Growing herbs as an organic crop offers new opportunities for farmers, and sometimes, especially in developing countries, opportunities for whole communities. In northeastern Brazil, for example, community-run herb gardens grow medicinal herbs that are sold to local hospitals. Doctors at the hospital then prescribe these medicines for their patients.

The rise in popularity of herbal medicines, however, also directly threatens the survival of some wild species. Demand for goldenseal (*Hydrastis canadensis*, p. 105) has become so great that it now fetches around $140 a pound (£170 a kilo). It was a common plant in the woodlands of northern America two centuries ago, but is now an endangered species, with its survival in the wild threatened by overcollection. This example is by no means unique, and, sadly, many species are similarly threatened across the planet. The extinction of plant species as a result of over-intensive collecting is nothing new. The herb silphion, a member of the carrot family, was used extensively as a contraceptive by the women of ancient Rome.